

NPTEL Course on
"Computer System Optimization for ML"

Lecture #21

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Agenda

- Recap
- Hybrid Parallelism

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Tensor Parallelism

- **Model parallelism, Pipeline Parallelism split layers across workers (GPUs)**
- **What if a layer itself is so large that a GPU cannot handle it?**
- **Tensor parallelism**
  - Split a tensor into N chunks along a specific dimension and each device only holds 1/N of the whole tensor while not affecting the correctness of the computation graph
  - This requires additional communication to make sure that the result is correct

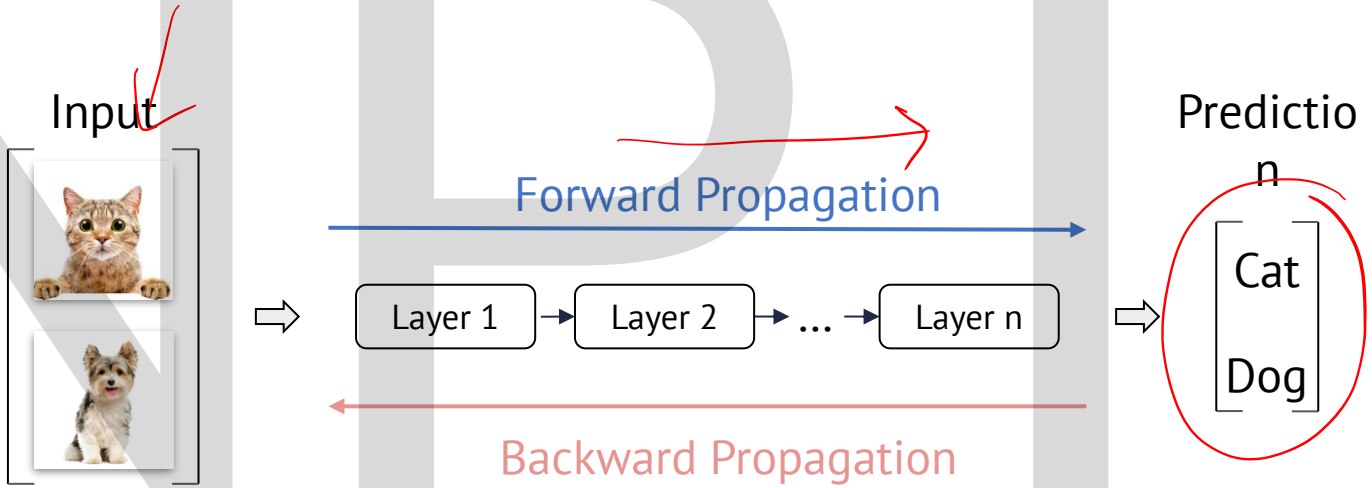
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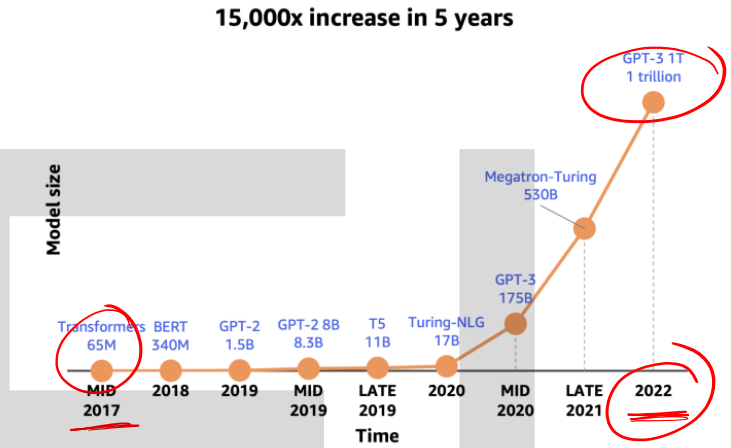


HYBRID PARALLELISM

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What are the System Challenges?





1. What if the **input dataset** is very large?

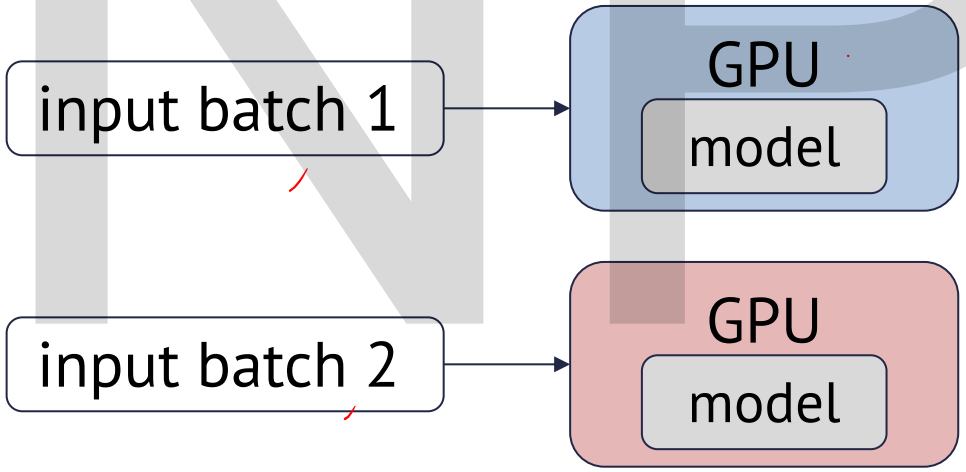
2. What if the **model** is very large?

What are the System Challenges?

1. What if the input dataset is very large?

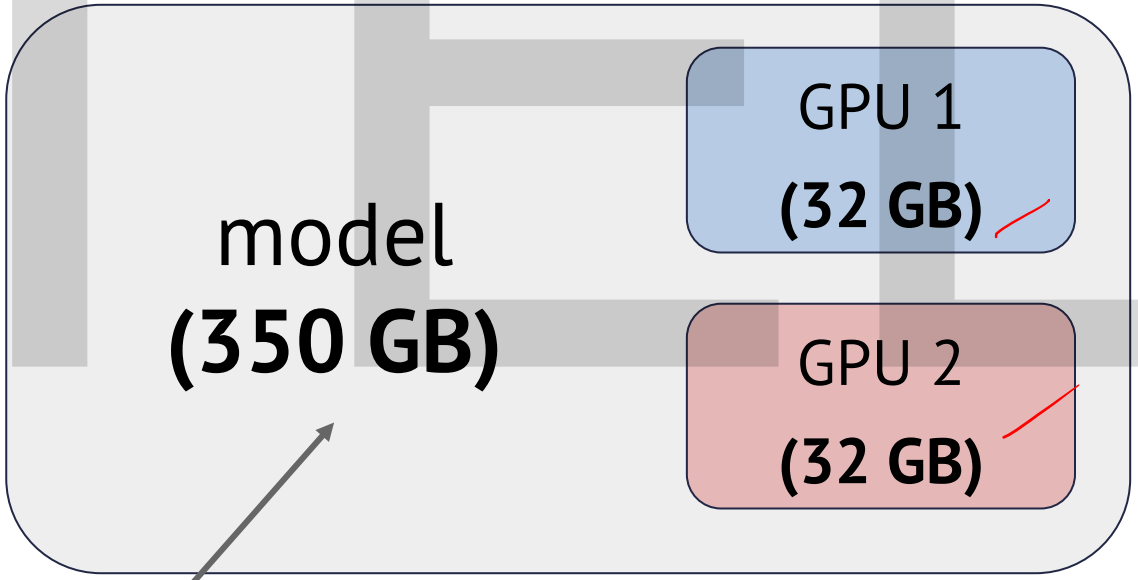
😃 **Easy.**

Use data parallelism: partition input data and replicate the model



2. What if the model is very large?

😖 **Hard !!**



Challenge: How to partition a computational graph?

Why is this Challenging?

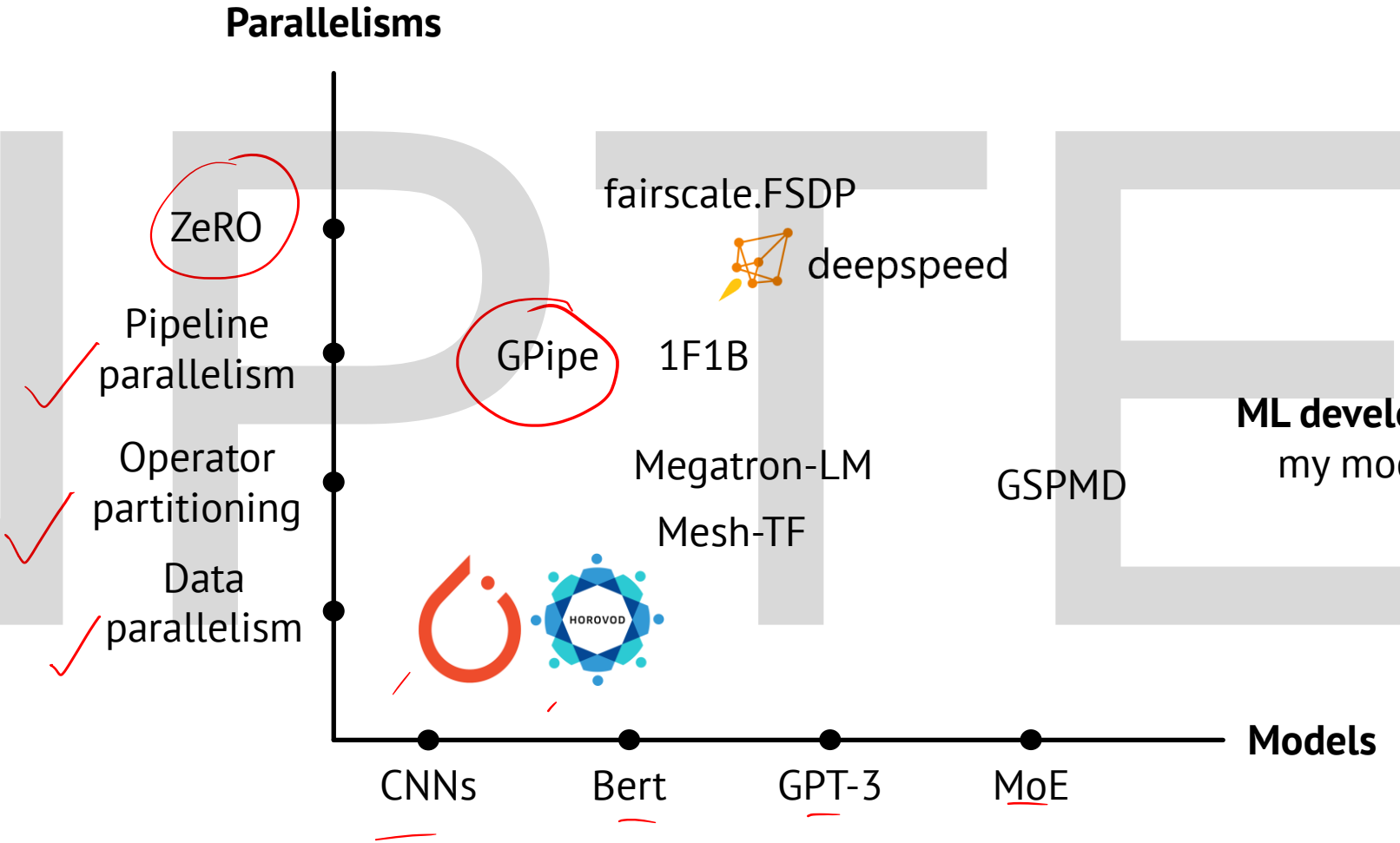
For a general computational graph we need to decide on:

- Which axes to break the model: operator vs pipeline
- How to efficiently map the GPUs to the resulting execution graph
- …

For a general non-linear graph this leads to a combinatorically large search space

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Auto-parallelization: Motivation





ML developer: which one is for my model and my cluster?

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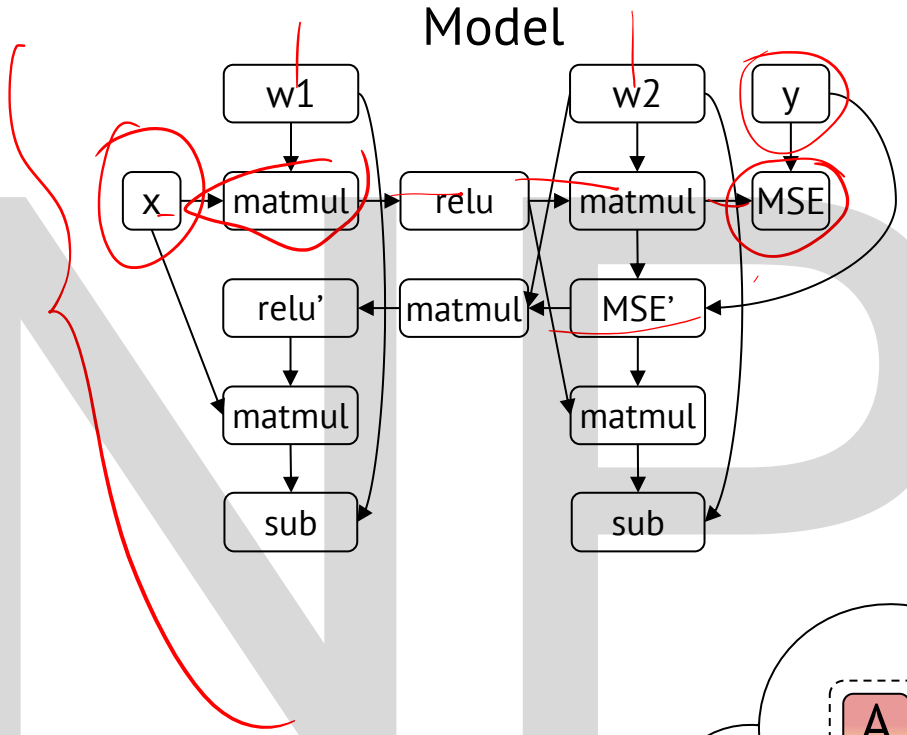
Auto-parallelization: Problem

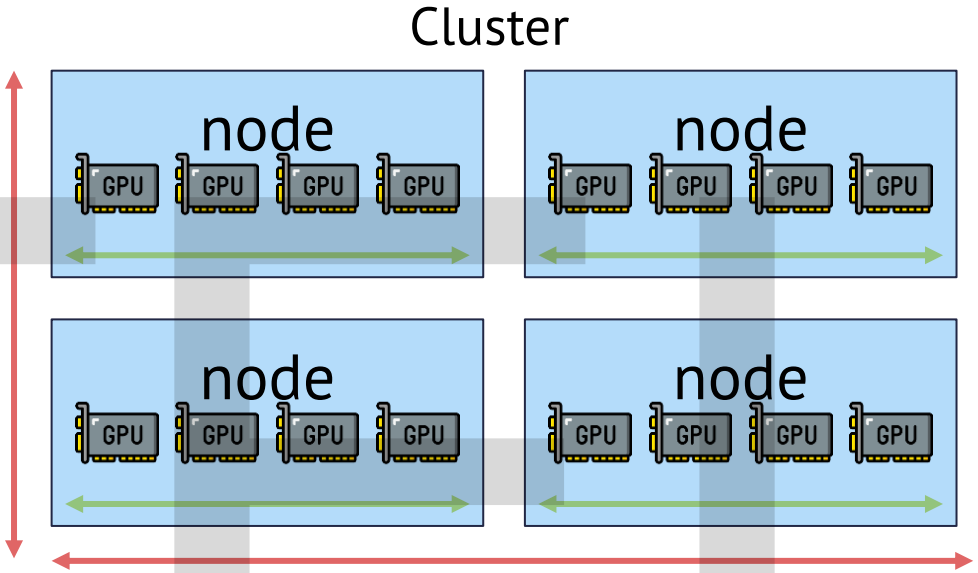
$$\max_{\text{strategy}} \ \text{Performance}(\text{Model}, \text{Cluster})$$

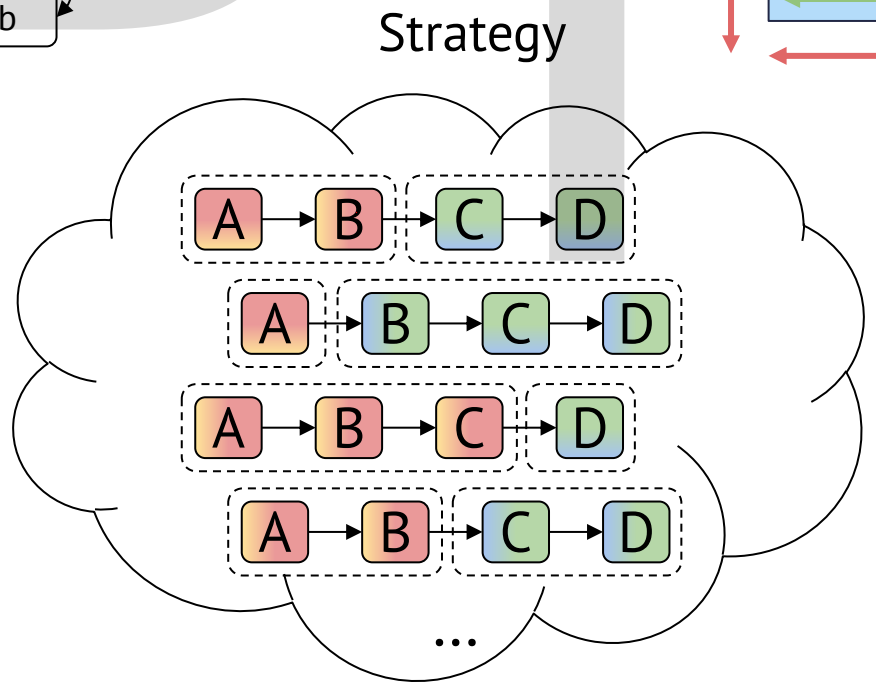
$$s.t. \ \text{strategy} \in \text{Inter-op} \cup \text{Intra-op}$$

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Auto-parallelization: Problem







The Search Space is Huge

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#ops in a real model (nodes to color)	#op types (type of nodes)	#devices on a cluster (available colors)
100 - 10K	80 - 200+	10s - 1000s

Automatic Parallelization Methods

Search-based methods

- MCMC:
  - [Jia et al., 2018]
  - **[Jia et al., 2019]**
- Heuristics
  - [Fan et al., 2021]

Learning-based methods

- Reinforcement Learning:
  - **[Mirhoseini et al., 2017]**
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  - [Addanki, et al., 2019]
- ML-based cost model:
  - [Chen et al., 2018],
  - [Zhou et al., 2020],
  - [Zhang, 2020]
- Bayesian optimization:
  - [Sergeev et al., 2018],
  - [Peng et al., 2019]

Optimization-based methods

- Dynamic programming
  - [Wang, et al., 2018]
  - [Narayanan, et al., 2019]
  - [Li, et al., 2021]
  - [Narayanan, et al., 2012]
  - [Tarnawski, et al., 2020]
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- Integer linear programming
  - [Tarnawski, et al., 2020]
- Hierarchical optimization
  - **[Zheng, et al., 2022]**

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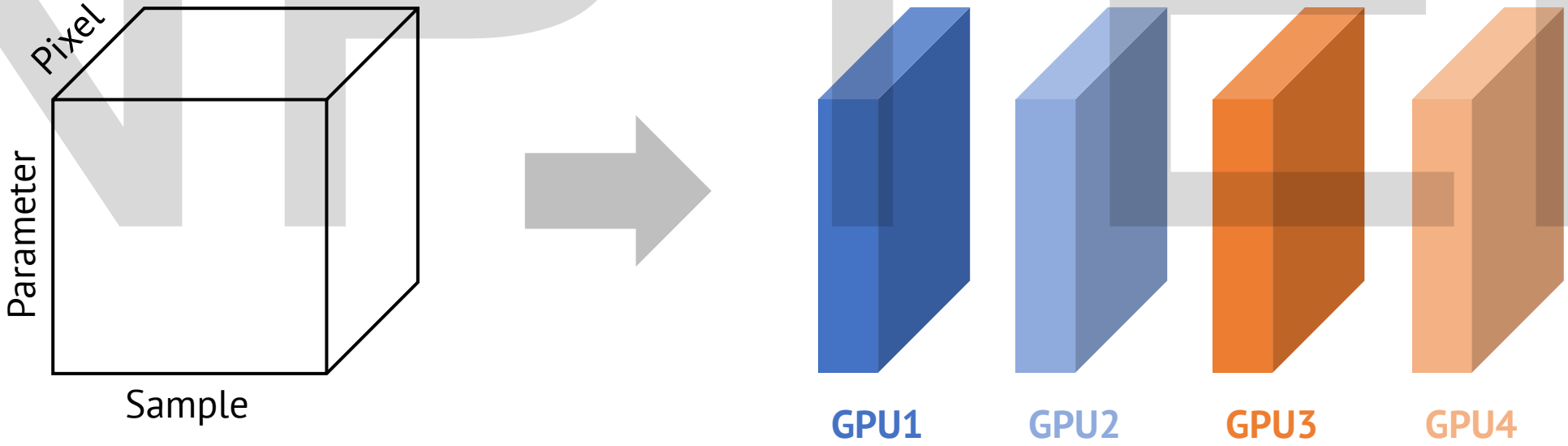
The SOAP Search Space

- Samples
- Operators
- Attributes
- Parameters

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The SOAP Search Space

- **Samples: partitioning training samples (data parallelism)**
- **Operators**
- **Attributes**
- **Parameters**

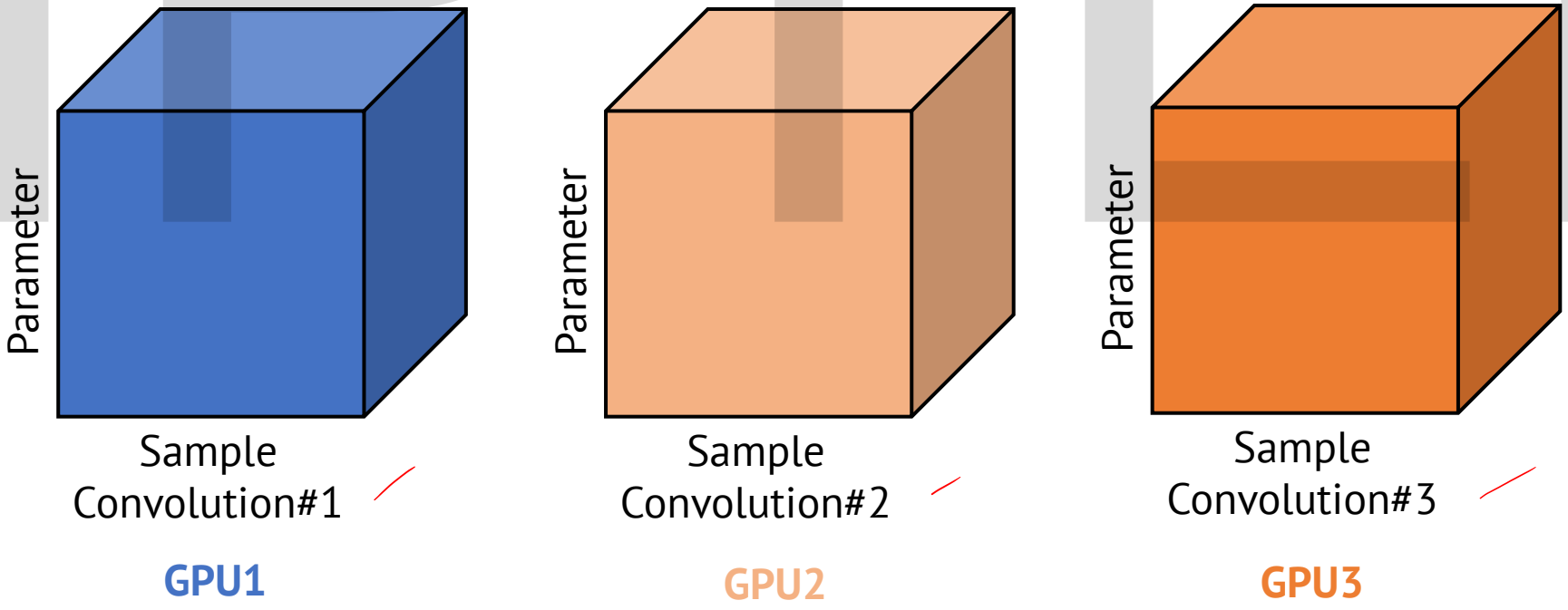


Parallelizing a 1D convolution

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The SOAP Search Space

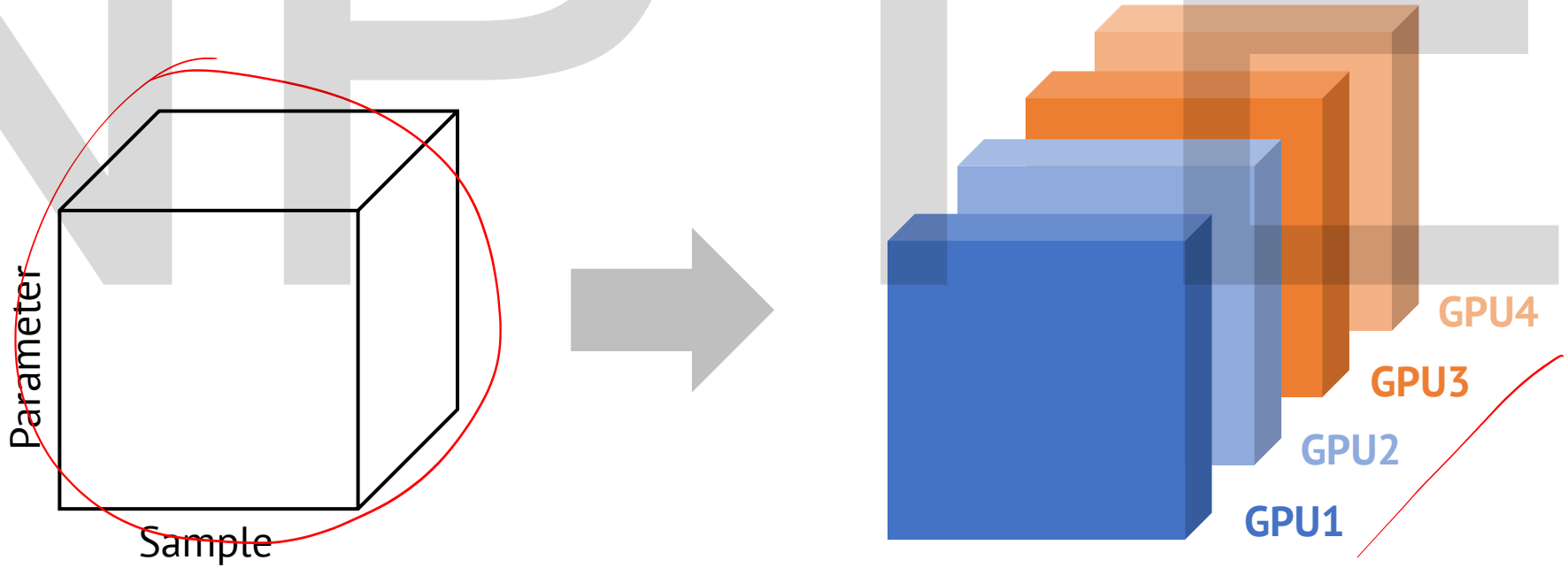
- **Samples:** partitioning training samples (data parallelism)
- **Operators: partitioning DNN operators**
- **Attributes**
- **Parameters**



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The SOAP Search Space

- **S**amples: partitioning training samples (data parallelism)
- **O**perators: partitioning DNN operators
- **A**ttributes: partitioning attributes in a sample (e.g., different pixels)
- **P**arameters

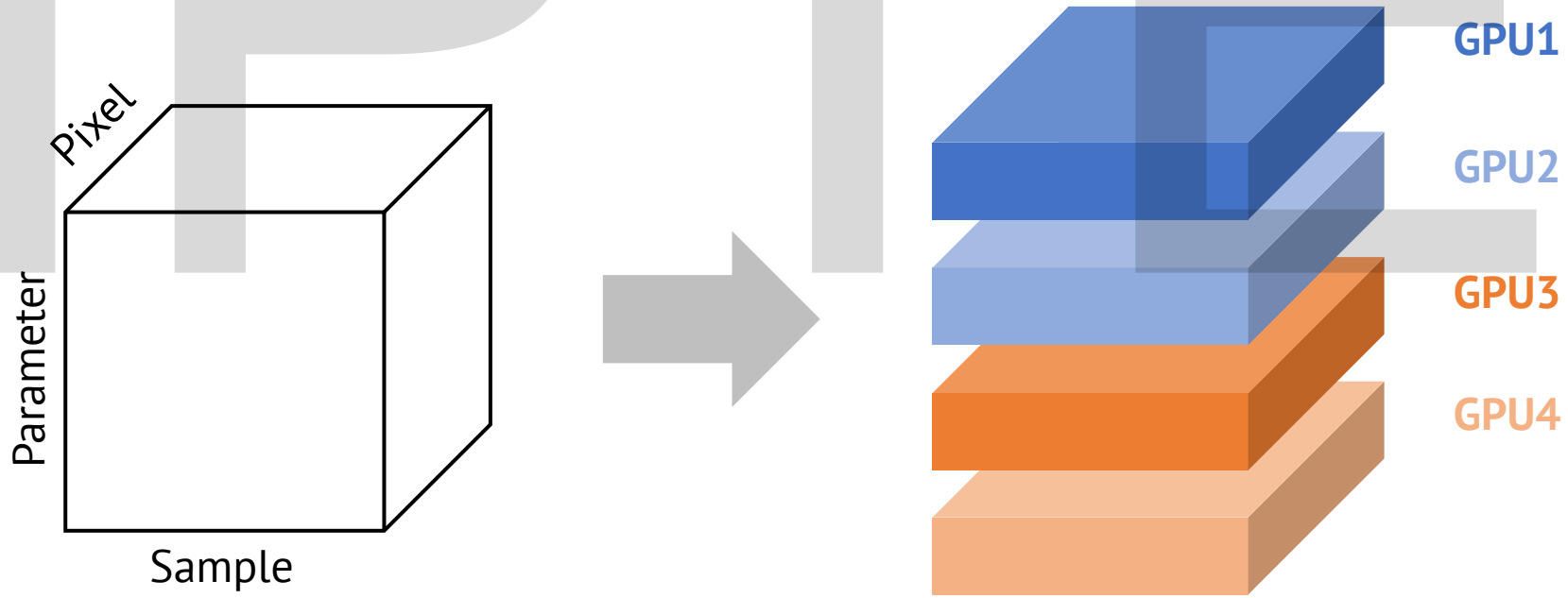


Parallelizing a 1D convolution

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The SOAP Search Space

- **Samples:** partitioning training samples (data parallelism)
- **Operators:** partitioning DNN operators
- **Attributes:** partitioning attributes in a sample (e.g., different pixels)
- **Parameters: partitioning parameters in an operator**



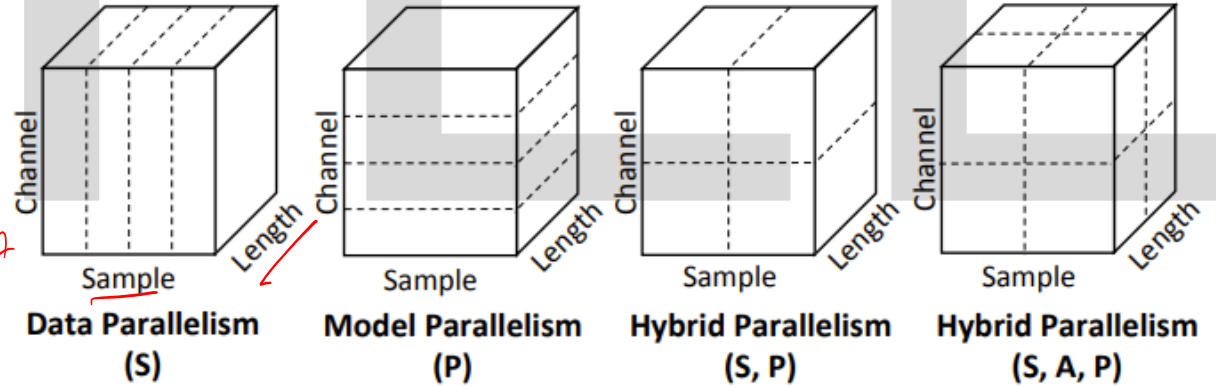
Parallelizing a 1D convolution

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FlexFlow: The Search Space

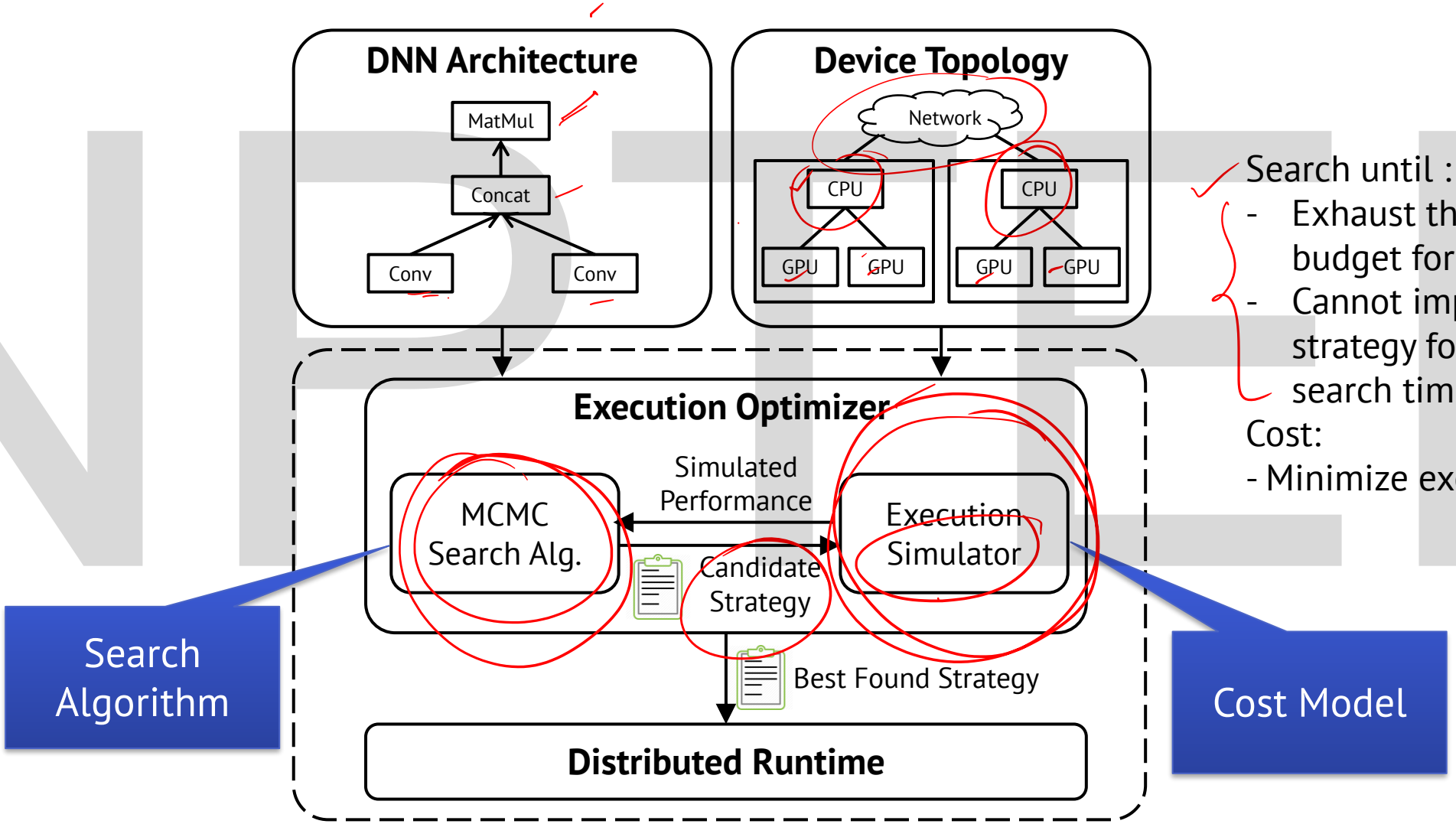
- SOAP Space
  - S (Sample) - sample dimension
  - O (Operation) – operator placement
  - P (Parameter) – split the parameter
  - A (Attribute) – the rest

Operator	Parallelizable Dimensions		
	(S)ample	(A)ttribute	(P)arameter
1D pooling	sample	length, channel	
1D convolution	sample	length	channel
2D convolution	sample	height, width	channel
Matrix multiplication	sample		channel



Jia, et al. "Beyond Data and Model Parallelism for Deep Neural Networks". *MLSys* 2019.

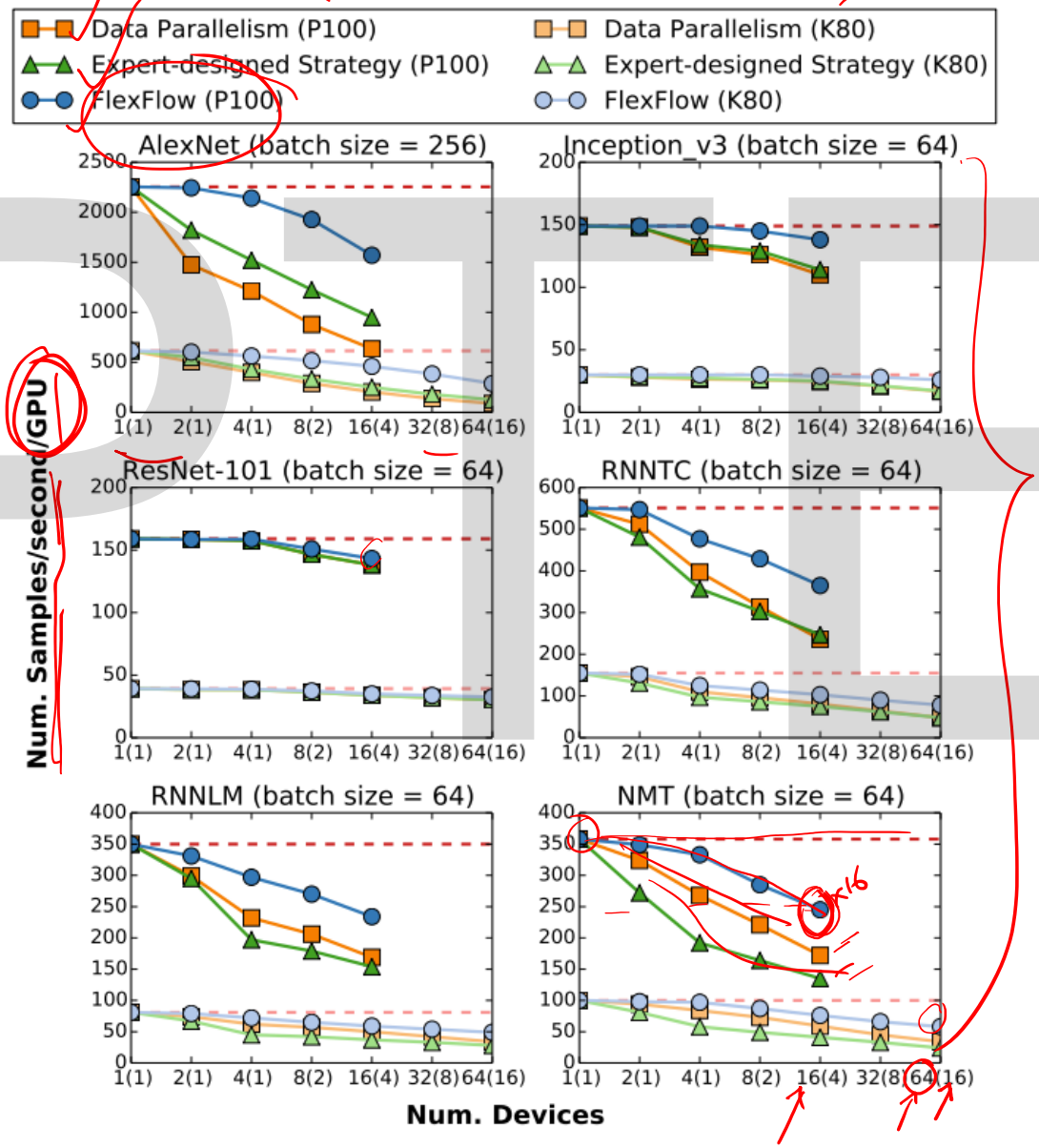
FlexFlow



Search until :
- Exhaust the time budget for search (OR)
- Cannot improve the strategy for half the search time

Cost:
- Minimize execution time

Results



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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)